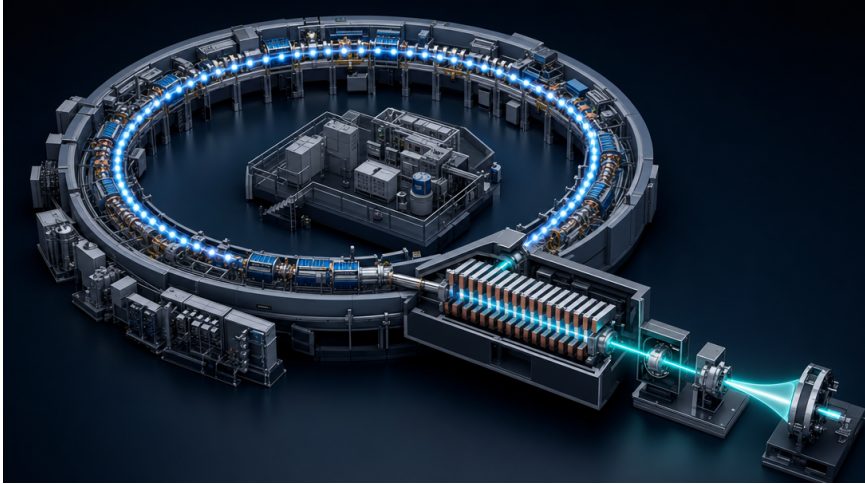


A Storage-Ring Steady-State-Microbunching Undulator Architecture for a High-Average-Power 13.5 nm Lithography Source

Architecture I of the lumen EUV source family (rank 2); shared experimental gate with
the compact SSMB-Compton architecture

lumen verified-hypothesis lab
dancinlab/lumen

Draft, as-of 2026-06



Storage-ring SSMB undulator source (conceptual illustration).

Abstract

We specify the *storage-ring steady-state-microbunching (SSMB) undulator* architecture for a continuous-wave, coherent 13.5 nm EUV source at the ~ 100 W-class average power high-volume lithography requires. A several-hundred-MeV electron beam stored in a ring is microbunched at the radiation wavelength every turn, so N electrons radiate *coherently* ($P \propto N^2$, a $\sim 10^6 \times$ enhancement) and, because the ring reuses the beam, the power is delivered continuous-wave rather than in the dumped single-shot of a single-pass free-electron laser (FEL). This is the missing middle between the incumbent laser-produced plasma and the funded conventional energy-recovery-linac FEL: coherent average power without a dumped beam. The mechanism is proof-of-principle demonstrated [1]; 13.5 nm kilowatt operation is undemonstrated. We give the closed-form architecture, its subsystems, and its binding gate — steady-state microbunching at 13.5 nm precision — which it *shares* with the compact SSMB-Compton architecture (companion paper II): one experiment de-risks both. The analysis is closed-form and reference-matched; it claims convergence on the same physics, not precedence.

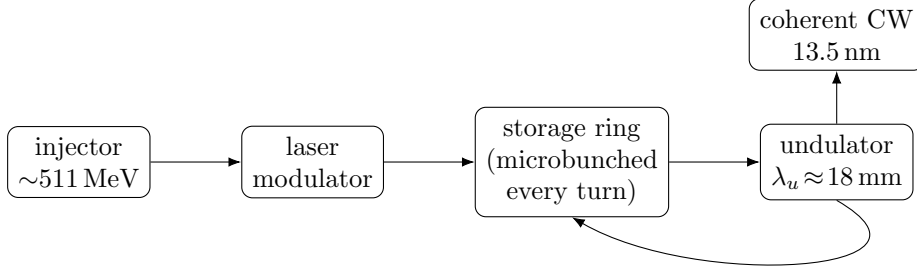


Figure 1: SSMB undulator architecture. The stored beam is laser-modulated and microbunched at the radiation wavelength every turn; on-axis undulator emission is coherent ($P \propto N^2$) and continuous-wave because the ring reuses the beam.

1 Introduction

This is Architecture I of the lumen EUV source family. For the EUV-litho high-flux requirement the family ranks eight source classes; this paper specifies rank 2, the SSMB undulator. It is one of two *novel* architectures (the other, rank 3, is the compact SSMB-Compton source of the companion paper). The two are genuinely different machines — different beam energy (~ 511 MeV vs. 7 MeV), different radiation mechanism (undulator vs. inverse-Compton), different footprint — so we document them separately, while making explicit that they reduce to one shared experimental gate (Section 4).

2 Background: the wall is flux, not wavelength

An undulator of period λ_u radiates at $\lambda = (\lambda_u/2\gamma^2)(1 + K^2/2)$, so wavelength is an electron-energy dial with no physical floor (H.011). The binding limit for an FEL is beam slice energy spread: gain collapses once $\sigma_\delta/\rho \gtrsim 1$ [2], and a representative 0.5% laser-plasma beam sits $\sim 5\times$ over budget, never lasing at 13.5 nm (H.031; demonstrated LPA-FEL floor ~ 25 nm [3]). A conventional RF-compressed beam clears this (FLASH lased 13 nm, 2006), which is why the funded near-term answer is a shared conventional ERL FEL (H.030). The architecture here makes flux — average power — the operative wall, and breaks it by coherence.

3 Architecture

In a storage ring N electrons normally radiate incoherently ($P \propto N$). Microbunched at the radiation wavelength ($\sigma_z \ll \lambda$) they radiate coherently ($P \propto N^2$, $\sim 10^6\times$). *Steady-state* microbunching sustains the bunching every turn, so the coherent power is continuous-wave from a reused beam (H.043). The program target is ~ 1 kW at 13 nm to 14 nm [4], $\geq 10\times$ the ~ 100 W HVM floor (H.005). Because the radiation is on-axis undulator emission, this architecture carries *no* Compton bandwidth-collection flux tax; its cost is the higher beam energy (~ 511 MeV at an 18 mm undulator) and hence a larger ring than the compact Compton variant.

Subsystems and secondary gates. Beyond the shared microbunching gate, the architecture-specific gates are: (i) the strong laser modulator synchronized to the ring; (ii) undulator radiation extraction; and (iii) the power-scaling enhancement/optics for kilowatt average power. Each is engineering, not forbidden.

4 The shared gate

The architecture rests on one unproven step: steady-state microbunching at 13.5 nm precision ($\sigma_z \approx 3$ nm). This is physics-allowed — the bunching factor $b = \exp(-\frac{1}{2}(2\pi\sigma_z/\lambda)^2) \approx 0.38$ gives a coherent fraction $b^2 \approx 0.14$ with no forbidding floor (H_046). **The same gate binds the compact SSMB-Compton architecture (paper II), so one experiment de-risks both** (H_048). The cooling lever the lab derived closed-form — open the beam-quality link by cooling, not by raising the Pierce ρ (H_024) — is exactly what the latest SSMB frontier adds: optical stochastic cooling combined with SSMB [5] (H_050).

5 Economics

The CAPEX win is amortization (M9): one $\sim \$260$ M / 10 kW source fanned out to ~ 10 scanner ports reaches $\sim \$26$ M/scanner ($\sim 8\times < \text{LPP}$) but inverts at fan-out 1 (H_031). The second-law waste-heat floor is $\sim 0.5\%$ of LPP cost — negligible (H_020). The cryomodule learning curve is shallow (Wright ~ 0.95), so volume manufacturing is not the lever.

6 Related work and limitations

The SSMB mechanism was demonstrated by Deng et al. [1]; the EUV-litho target is developed in Deng et al. [4], the cooling-enhanced variant in Deng [5]. All relations here are first-order, closed-form, representative-parameter — not start-to-end simulations. “Physics-allowed” is necessary, not sufficient: collective effects, ring nonlinearity, and jitter can erode σ_z . 13.5 nm kilowatt SSMB is undemonstrated; this is reference-match, not priority.

7 Reproducibility and future work

Closed-form evidence: `lumen HYPOTHESES/cards/H_{011,024,030,031,043,046,048,050}.md` + deterministic runners in `state/h0*/`; re-verify with `python3 tool/qa.py`. The decisive next step is experimental: demonstrate steady-state microbunching at 13.5 nm with $b^2 \gtrsim 0.1$ held over $\gtrsim 10^3$ turns; if σ_z degrades to $\gtrsim 10$ nm, the path reverts to longer wavelengths and the funded conventional ERL FEL remains the near-term answer.

References

- [1] Xiujie Deng et al. Experimental demonstration of the mechanism of steady-state microbunching. *Nature*, 590:576–579, 2021.
- [2] Ming Xie. Design optimization for an x-ray free electron laser driven by slac linac. *Proc. PAC*, 1995.
- [3] Hengyuan Xiao et al. Staged laser wakefield acceleration for saturated lasing of bandwidth-tunable free-electron lasers from euv to x-ray. *arXiv preprint arXiv:2603.28214*, 2026.
- [4] Xiujie Deng et al. A perspective of the kilowatt-level steady-state microbunching euv light source. *Scientific Reports*, 2022. s41598-022-07323-z; arXiv:2110.08987.
- [5] Xiujie Deng. Stochastic cooling enhanced steady-state microbunching. *arXiv preprint arXiv:2512.09399*, 2025.